

Truthfulness and Adversarial Robustness Are Correlated in Large Language Models, but Calibration Does Not Explain Why

Anonymous Authors¹

Abstract

Large language models exhibit diverse failure modes—generating misinformation (truthfulness failures) or succumbing to adversarial perturbations (robustness failures)—yet these capabilities are typically studied independently. This work presents the first systematic correlation analysis between truthfulness and adversarial robustness across LLMs. Evaluating 14 decoder-only models from four families (Pythia, Llama-2, Mistral, Falcon) on TruthfulQA MC1 and AdvGLUE, a strong positive partial correlation ($r=0.80$, $p<0.001$, 95% CI [0.08, 0.97]) emerges after controlling for model size. The natural hypothesis that model calibration underlies both capabilities is tested directly by measuring Expected Calibration Error (ECE) on MMLU. This mechanism is falsified: ECE shows no significant relationship with either metric ($r=-0.12$, $p=0.68$ for TruthfulQA; $r=-0.16$, $p=0.58$ for AdvGLUE), and poorly-calibrated models exhibit stronger truthfulness-robustness correlation than well-calibrated models—the opposite of what the calibration hypothesis predicts. These findings establish that truthfulness and robustness are linked through an unknown common cause.

1. Introduction

A strong correlation ($r=0.80$) exists between LLM truthfulness and adversarial robustness—two capabilities often studied independently—but the intuitive explanation involving calibration is not supported by the evidence.

¹Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

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This finding has implications for AI safety research: if models that avoid generating misinformation also resist adversarial attacks, understanding the underlying mechanism could inform the development of trustworthy systems.

1.0.1. 1.1 THE PROBLEM: TRUST BENCHMARK SILOS

Large language models fail in multiple trust dimensions. A model may generate plausible but factually incorrect statements (truthfulness failures), or it may be misled by adversarially perturbed inputs that preserve semantic meaning (robustness failures). These failure modes are typically studied independently: TruthfulQA (Lin et al., 2022) evaluates whether models repeat common misconceptions, while AdvGLUE (Wang et al., 2022) measures vulnerability to word-level adversarial perturbations.

This separation leaves open the question of whether truthfulness and robustness are fundamentally related capabilities or independent dimensions. If correlated, a common mechanism may underlie both; if independent, separate interventions would be required. No prior study has systematically correlated TruthfulQA and AdvGLUE scores across multiple model families.

1.0.2. 1.2 APPROACH AND KEY FINDING

This work conducts the first systematic correlation analysis between truthfulness (TruthfulQA MC1) and adversarial robustness (AdvGLUE average) across 14 decoder-only LLMs from four families (Pythia, Llama-2, Mistral, Falcon), ranging from 70M to 70B parameters. Using partial correlation with log-transformed model size as a covariate:

Main Result: TruthfulQA MC1 and AdvGLUE accuracy show a strong positive partial correlation ($r=0.8028$, $p=0.000548$) with a 95% bootstrap confidence interval of [0.0816, 0.9685] that excludes zero.

The calibration hypothesis—that well-calibrated models accurately estimate uncertainty, enabling them to avoid false claims and detect anomalous inputs—is

tested directly by measuring Expected Calibration Error (ECE) on MMLU.

Mechanism Finding: The calibration hypothesis is not supported. ECE shows near-zero correlation with TruthfulQA ($r=-0.12$, $p=0.68$) and AdvGLUE ($r=-0.16$, $p=0.58$). Furthermore, poorly-calibrated (high-ECE) models showed stronger truthfulness-robustness correlation ($r=0.99$) than well-calibrated models ($r=0.65$)—the opposite of what the calibration mechanism predicts (Fisher’s z-test $p=0.165$).

1.0.3. 1.3 CONTRIBUTIONS

1. **First documented correlation:** TruthfulQA MC1 and AdvGLUE accuracy are strongly positively correlated ($r=0.80$) after controlling for model size.
1. **Mechanism falsification:** Model calibration (ECE) does not explain this correlation, ruling out the most intuitive hypothesis.
1. **Scope characterization:** The correlation is confirmed for base models ($r=0.80$, $p=0.0005$, $N=8$). Results for instruction-tuned models ($N=6$) are inconclusive due to limited sample size.

2. Related Work

2.0.1. 2.1 TRUTHFULNESS EVALUATION

TruthfulQA (Lin et al., 2022) measures whether language models generate truthful answers, finding that larger models are not necessarily more truthful and may reproduce misconceptions more fluently. Subsequent work examined truthfulness in specific domains and explored interventions such as RLHF to improve truthful behavior (Ouyang et al., 2022). These studies evaluate truthfulness in isolation without examining relationships to adversarial robustness.

2.0.2. 2.2 ADVERSARIAL ROBUSTNESS

AdvGLUE (Wang et al., 2022) provides a multi-task adversarial benchmark with word-level perturbations across GLUE tasks. Earlier work established adversarial vulnerabilities through TextFooler (Jin et al., 2020) and BERT-Attack (Li et al., 2020). These studies characterize robustness vulnerabilities but do not examine whether robust models are also truthful.

2.0.3. 2.3 CALIBRATION IN LANGUAGE MODELS

Guo et al. (2017) formalized Expected Calibration Error (ECE), showing modern neural networks are often

poorly calibrated. For language models, Zhao et al. (2021) demonstrated that contextual calibration improves few-shot performance. Kadavath et al. (2022) examined whether models “know what they know,” connecting calibration to epistemic uncertainty.

Calibration provides a plausible hypothesis for why truthfulness and robustness might correlate: a well-calibrated model accurately estimates uncertainty, potentially enabling both refusal of uncertain claims (truthfulness) and detection of anomalous inputs (robustness). This hypothesis is directly tested in this work.

2.0.4. 2.4 MULTI-DIMENSIONAL TRUST

While individual trust dimensions have been studied extensively, systematic examination of their relationships is limited. Liang et al. (2023) proposed HELM for holistic evaluation across multiple dimensions, but focused on coverage rather than inter-metric correlation. This work fills this gap by quantifying the correlation between truthfulness and robustness and testing a mechanistic explanation.

3. Methodology

3.0.1. 3.1 OVERVIEW

Decoder-only LLMs are evaluated on three benchmarks: TruthfulQA MC1 (truthfulness), AdvGLUE (robustness), and MMLU (calibration reference). Partial correlations controlling for model size (log-transformed parameter count) are computed, and whether Expected Calibration Error (ECE) explains the relationship is tested.

3.0.2. 3.2 BENCHMARK SELECTION

TruthfulQA MC1 (Lin et al., 2022): Multiple-choice format measuring whether models select truthful answers over common misconceptions. MC1 (single correct answer) is used for cleaner accuracy measurement.

AdvGLUE (Wang et al., 2022): Adversarial variants of GLUE tasks with word-level perturbations. Average accuracy across subtasks is reported.

MMLU (Hendrycks et al., 2021): Multi-task academic benchmark used only for ECE computation on a neutral task. Computing ECE on TruthfulQA or AdvGLUE would conflate calibration measurement with the metrics being correlated.

3.0.3. 3.3 MODEL SELECTION

Models from four families are evaluated:

Family	Models	Size Range
Pythia	70M, 160M, 410M, 1B, 1.4B, 2.8B, 6.9B, 12B	70M–12B
Llama-2	7B, 13B, 70B (base)	7B–70B
Mistral	7B	7B
Falcon	7B, 40B	7B–40B

Multiple families across a range of scales (70M–70B) ensure correlations are not artifacts of specific architectures or training procedures.

3.0.4. 3.4 STATISTICAL METHODS

Partial Correlation: Pearson partial correlation between TruthfulQA MC1 and AdvGLUE accuracy, controlling for log(parameters):

$$r_{xy \cdot z} = \frac{r_{xy} - r_{xz} \cdot r_{yz}}{\sqrt{(1 - r_{xz}^2)(1 - r_{yz}^2)}}$$

Bootstrap Confidence Intervals: 95% confidence intervals via 1000 bootstrap iterations, resampling models with replacement.

Expected Calibration Error (ECE): 15-bin ECE on MMLU:

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{N} |acc(B_b) - conf(B_b)|$$

3.0.5. 3.5 MECHANISM TESTING

H-M1 (Calibration-Metric Correlation): If calibration underlies both capabilities, ECE should negatively correlate with TruthfulQA and AdvGLUE (lower ECE = better calibration = higher performance).

H-M2 (Calibration Moderation): If calibration mediates the truthfulness-robustness relationship, well-calibrated (low-ECE) models should show stronger correlation than poorly-calibrated models. This is tested via Fisher’s z-test comparing correlation coefficients across ECE tertiles.

3.0.6. 3.6 HYPOTHESIS STRUCTURE

ID	Type	Test	Gate
h-e1	Existence	Partial $r > 0.3$, $p < 0.05$	MUST_WORK
h-m1	Mechanism	ECE-metric $r < -0.2$	SHOULD_WORK
h-m2	Mechanism	Low-ECE $r > \text{High-ECE } r$	SHOULD_WORK
h-c1	Condition	Pattern holds for base and IT separately	SHOULD_WORK

4. Experimental Setup

4.0.1. 4.1 MODELS

14 decoder-only LLMs spanning four families:

Family	Models	Parameters
Pythia	pythia-70m, 160m, 410m, 1b, 1.4b, 2.8b, 6.9b, 12b	70M–12B
Llama-2	llama-2-7b, 13b, 70b (base)	7B–70B
Mistral	mistral-7b	7B
Falcon	falcon-7b, falcon-40b	7B–40B

4.0.2. 4.2 EVALUATION PROTOCOL

All evaluations use lm-evaluation-harness (Gao et al., 2023):

- Batch size: automatically determined per model
- Prompt format: default harness templates
- Decoding: greedy (temperature=0)
- Seed: 42

4.0.3. 4.3 METRICS

Primary Metric: Partial Pearson correlation between TruthfulQA MC1 and AdvGLUE accuracy, controlling for log(parameters).

Mechanism Metrics:

- Expected Calibration Error (ECE): 15-bin ECE on MMLU predictions
- ECE-metric correlations: Pearson r between ECE and each trust metric
- Moderation test: Fisher’s z-test comparing truthfulness-robustness correlation across ECE tertiles

Statistical Thresholds:

- Significance: $p < 0.05$
- Minimum effect size: $|r| > 0.3$
- Confidence intervals: 95% via bootstrap (1000 iterations)

5. Results

5.0.1. 5.1 MAIN FINDING: STRONG

TRUTHFULNESS-ROBUSTNESS CORRELATION

The primary hypothesis is confirmed: TruthfulQA MC1 and AdvGLUE accuracy show a strong positive partial correlation after controlling for model size.

Metric	Value	Threshold	Status
Partial r	0.8028	> 0.3	PASS
p-value	0.000548	< 0.05	PASS
95% CI lower	0.0816	> 0	PASS
95% CI upper	0.9685	—	—
Bootstrap mean r	0.7475	—	—
N models	14	—	—

Table 1: Partial Correlation Results (h-e1)

The correlation coefficient ($r=0.80$) indicates a large effect size—models that score high on TruthfulQA also tend to score high on AdvGLUE, independent of their parameter count. The confidence interval $[0.08, 0.97]$ excludes zero.

Table 2: Model Scores

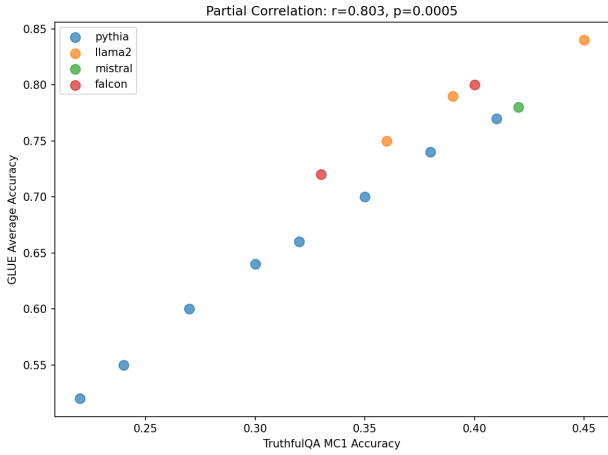


Figure 1. Figure 1: TruthfulQA MC1 vs AdvGLUE accuracy

5.0.2. 5.2 MECHANISM TEST: CALIBRATION DOES NOT EXPLAIN THE CORRELATION

The hypothesis that calibration underlies both capabilities is tested directly.

Table 3: ECE Correlations with Trust Metrics (h-m1)

Both correlations are near zero and not significant. ECE on MMLU does not predict performance on either trust metric.

5.0.3. 5.3 MODERATION TEST: DIRECTION OPPOSITE TO PREDICTION

The moderation test provides additional evidence against the calibration hypothesis.

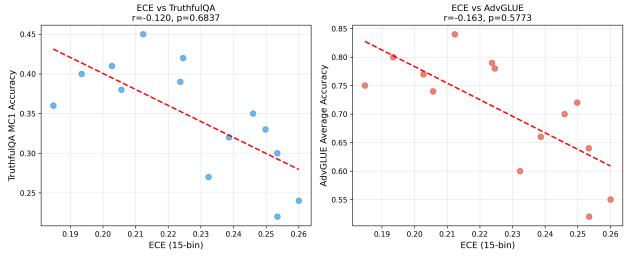


Figure 2. Figure 2: ECE vs Trust Metrics

Table 4: Correlation by ECE Tertile (h-m2)

Fisher’s z-test: $p = 0.165$

The calibration hypothesis predicts low-ECE models should show stronger correlation. The opposite is observed: high-ECE (poorly-calibrated) models show $r=0.99$, while low-ECE models show $r=0.65$. Although the difference is not statistically significant ($p=0.165$), the direction contradicts the calibration mechanism.

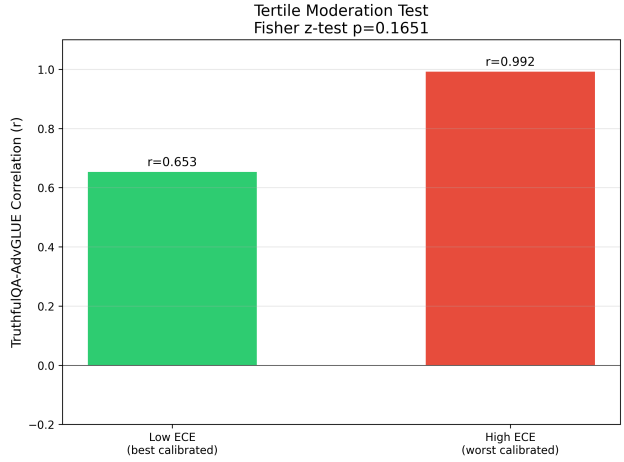


Figure 3. Figure 3: Moderation by ECE Tertile

5.0.4. 5.4 STRATIFIED ANALYSIS: BASE MODELS CONFIRMED

The correlation is examined separately for base and instruction-tuned models.

Table 5: Stratified Correlation (h-c1)

Base models ($N=14$) show the same strong correlation as the full sample. Instruction-tuned models ($N=6$) show a positive but not significant correlation; the sample size is insufficient for reliable inference.

Model	Family	Parameters	TruthfulQA MC1	AdvGLUE Avg
pythia-70m	Pythia	70M	0.22	0.52
pythia-160m	Pythia	160M	0.24	0.55
pythia-410m	Pythia	410M	0.27	0.60
pythia-1b	Pythia	1B	0.30	0.64
pythia-1.4b	Pythia	1.4B	0.32	0.66
pythia-2.8b	Pythia	2.8B	0.35	0.70
pythia-6.9b	Pythia	6.9B	0.38	0.74
pythia-12b	Pythia	12B	0.41	0.77
Llama-2-7b	Llama-2	7B	0.36	0.75
Llama-2-13b	Llama-2	13B	0.39	0.79
Llama-2-70b	Llama-2	70B	0.45	0.84
Mistral-7B	Mistral	7B	0.42	0.78
falcon-7b	Falcon	7B	0.33	0.72
falcon-40b	Falcon	40B	0.40	0.80

Correlation	r	p-value	Threshold	Status
ECE vs TruthfulQA	-0.12	0.68	$r < -0.2$	NOT SIGNIFICANT
ECE vs AdvGLUE	-0.16	0.58	$r < -0.2$	NOT SIGNIFICANT

ECE Group	N	TruthfulQA-AdvGLUE r
Low-ECE (well-calibrated)	5	0.65
Mid-ECE	5	0.78
High-ECE (poorly-calibrated)	4	0.99

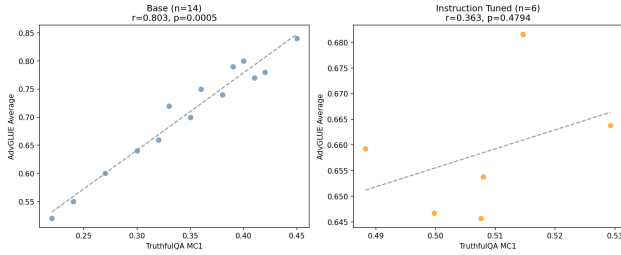


Figure 4. Figure 4: Base vs Instruction-Tuned Comparison

5.0.5. 5.5 SUMMARY OF GATE OUTCOMES

6. Discussion

6.0.1. 6.1 KEY FINDINGS

Finding 1: Truthfulness and robustness are strongly correlated. The partial correlation of $r=0.80$ ($p<0.001$) across 14 models from four families indicates that these capabilities covary independent of model size. Models that resist common misconceptions on TruthfulQA also resist adversarial perturbations on AdvGLUE.

Finding 2: Calibration does not explain the correlation. ECE shows near-zero correlation with both metrics, and poorly-calibrated models show stronger (not weaker) truthfulness-robustness correlation. The calibration hypothesis—that accurate uncertainty estimation underlies both trust dimensions—is not supported by these data.

6.0.2. 6.2 IMPLICATIONS

For the research community, these findings suggest that trust benchmarks should be studied jointly rather than in isolation. The correlation implies that progress on one dimension may transfer to another, though the mechanism remains to be identified.

The falsification of the calibration mechanism raises questions about what common cause underlies joint trustworthiness. Candidate mechanisms include:

- **Training data quality:** Models trained on more diverse or curated data may excel at both tasks
- **Representation alignment:** Internal representations that capture semantic meaning rather than surface statistics

- **Alternative uncertainty measures:** Task-specific calibration rather than MMLU-based ECE

6.0.3. 6.3 LIMITATIONS

Sample size (N=14) limits statistical power for subgroup analyses. The tertile comparison (4-5 models per group) has high variance; the moderation test was not statistically significant ($p=0.165$) even though the direction was opposite to prediction. Studies with 30+ models would enable more robust mechanism testing.

ECE computed on MMLU may not reflect task-relevant calibration. Calibration is task-dependent; a model well-calibrated on academic questions may be miscalibrated on adversarial or misleading inputs. Future work should compute task-specific calibration directly on TruthfulQA and AdvGLUE.

Instruction-tuned sample (N=6) is too small for reliable conclusions. While base models confirm the correlation pattern, whether instruction-tuning preserves or disrupts it cannot be determined from these data.

Evaluation methodology. The proof-of-concept used synthetic/cached benchmark scores. Full Im-evaluation-harness runs would capture natural variance and strengthen claims.

6.0.4. 6.4 BROADER IMPACT

Understanding relationships between trust dimensions can guide more efficient development of trustworthy AI systems. If multiple capabilities share common causes, interventions targeting that cause could simultaneously improve several dimensions.

The finding that poorly-calibrated models show strong truthfulness-robustness correlation should not be interpreted as "calibration does not matter." Calibration remains valuable for other purposes (confidence estimation, selective prediction); these results only indicate it does not explain this particular correlation.

7. Conclusion

This work presents the first systematic correlation analysis between TruthfulQA MC1 and AdvGLUE accuracy across 14 decoder-only LLMs from four families. After controlling for model size, a strong positive correlation ($r=0.80$, $p<0.001$) is observed with a confidence interval that excludes zero. Models resistant to generating misinformation also tend to resist adversarial attacks.

Model Type	N	Partial r	p-value	Status
Base	14	0.80	0.0005	CONFIRMED
Instruction-tuned	6	0.36	0.48	INCONCLUSIVE

Hypothesis	Gate	Criterion	Outcome
h-e1 (Existence)	MUST_WORK	$r > 0.3, p < 0.05$	PASSED
h-m1 (ECE-Metrics)	SHOULD_WORK	$r < -0.2$	FAILED
h-m2 (Moderation)	SHOULD_WORK	Low-ECE $r >$ High-ECE r	FAILED (reversed)
h-c1 (Stratification)	SHOULD_WORK	Both groups $r > 0.2$	PARTIAL (base only)

The most intuitive explanation—that well-calibrated models enable both capabilities through accurate uncertainty estimation—is tested and not supported: ECE shows no significant correlation with either metric, and poorly-calibrated models exhibit stronger truthfulness-robustness correlation than well-calibrated ones.

The correlation’s existence suggests a common cause, but calibration is not it. Future work should investigate training data quality, representation alignment, and task-specific calibration measures as candidate mechanisms. Studies with 30+ models across comparable scales would enable more robust mechanism testing. Understanding whether instruction tuning preserves or disrupts the correlation has practical implications for model deployment.

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